

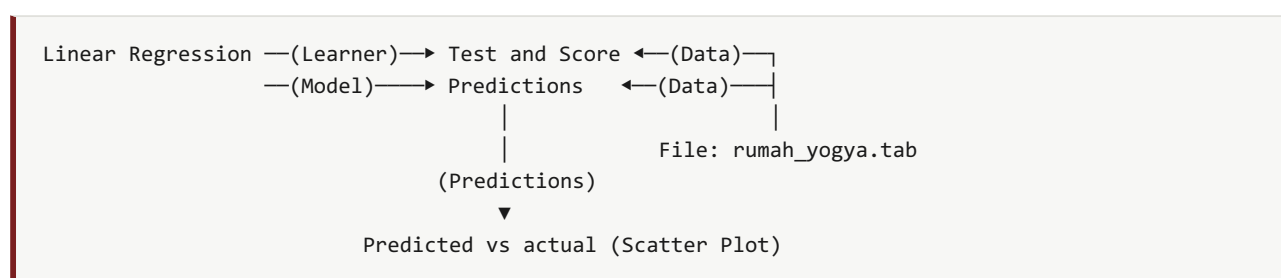
Week 6 — week06_regression.ows

Companion notes for the Orange workflow. Book: chapter 6. Notebook twin: `notebooks/week06_regression.ipynb`.

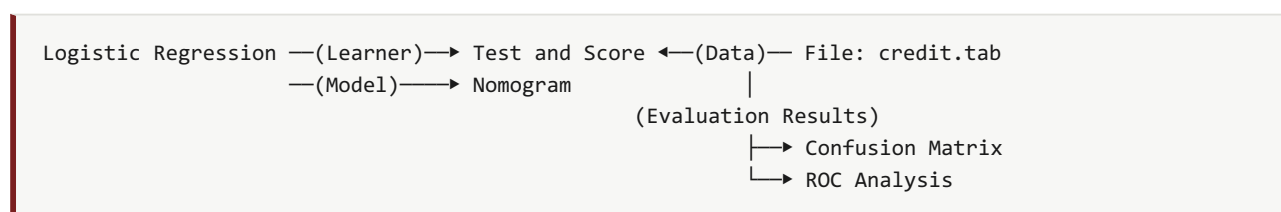
1. What this workflow is

Two independent pipelines on one canvas, deliberately laid out as two rows so you can see that predicting a number and predicting a class use the same skeleton with different parts.

Top row — continuous target (`harga`):



Bottom row — binary target (`default`):



Datasets — `data/rumah_yogya.tab` (2,200 houses, target `harga`) and `data/credit.tab` (4,000 applications, target `default`).

The idea being taught: three distinct channel types. This is the workflow where Orange's type system becomes visible.

Channel	Carries	Goes from → to
<code>Data</code>	a table of rows	File → everything
<code>Learner</code>	an <i>untrained</i> algorithm	Linear/Logistic Regression → Test and Score
<code>Model</code>	a <i>trained</i> predictor	Linear/Logistic Regression → Predictions, Nomogram
<code>Evaluation Results</code>	per-fold predictions + truth	Test and Score → Confusion Matrix, ROC

A model widget emits both `Learner` and `Model` , and which one you connect decides what happens. Send the `Learner` to Test and Score and it is re-fitted inside every fold — honest. Send the `Model` and it was fitted on everything, including the rows it is about to be tested on — leakage. **The wire you choose is the difference between a valid score and a fraudulent one.**

The Nomogram is the payoff widget: it turns a logistic regression into something a loan officer can read without knowing what a log-odds is.

2. What to do

1. **Test and Score (top)** — set *Cross validation, 5 folds*. Note RMSE, MAE, R^2 .
 2. **Predicted vs actual** — is the spread even across the range, or does it fan out? That fan is heteroscedasticity (book §6.4.1).
 3. **Logistic Regression** — try Ridge (L2) with $C = 1$, then Lasso (L1). Watch coefficients vanish.
 4. **Nomogram** — each feature is a line; the length of its bar is its influence. Drag the blue markers to score an applicant.
 5. **Confusion Matrix** — which error does the model make more often? Is that the cheap one?
-

Versi mahasiswa. Pembahasan, hal yang perlu diperhatikan, dan jawaban setiap langkah disampaikan di kelas setelah praktikum.